

Integer 4-Flow Reconfiguration on Odd Prism Graphs

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Abstract

Let $P_n = C_n \square K_2$ be the prism graph. Fix an orientation, and form the reconfiguration graph whose vertices are the integer nowhere-zero 4-flows on P_n and in which two distinct flows are adjacent when the support of their difference is a single cycle. We prove that this graph is nonempty and connected for every odd $n \geq 5$. The key step is an edge-lifting lemma: on every odd prism, each adjacency of nowhere-zero $\mathbb{Z}/4\mathbb{Z}$ -flows has adjacent integer representatives with the same support. Modular prism connectivity proved by Cranston, Li, Su, Wang, and Xu, together with the residue-fiber lemma of Esperet, Hendrey, Lagoutte, Marseloo, Norin, and Steiner, then gives the result. The proof is constructive and treats the sign-sensitive boundary case explicitly. Exact finite computations are reported only as corroborating checks, not as the proof of the statement for all n .

1 Introduction

For a fixed orientation of a graph, an integer nowhere-zero k -flow is a conserved integer edge function f satisfying $0 < |f(e)| < k$ on every edge. Flow reconfiguration allows one to change a flow along a single cycle while retaining these constraints. Esperet et al. introduced this reconfiguration problem in both the group-valued and integer settings [2]. They observed that integer connectivity implies modular connectivity and asked about the converse; they also proved that integer flows with the same residues lie in one reconfiguration component [2, Lemma 5.5].

Cranston et al. proved that every prism P_n with $n \geq 4$ has a connected nowhere-zero $\mathbb{Z}/4\mathbb{Z}$ -flow reconfiguration graph [1, Theorem 3.4]. For odd n , modular connectivity does not by itself control whether a modular cycle change can be represented by integers in $\{-3, -2, -1, 1, 2, 3\}$. The additional ingredient here is a uniform representative choice for every modular edge. Its delicate case is a residue change by 2: the representatives 1 and -1 inside the changed interval make both boundary spokes remain nonzero and bounded.

Our main conclusion is the following.

Theorem 1.1. *For every odd integer $n \geq 5$ and every fixed orientation of $P_n = C_n \square K_2$, the reconfiguration graph of integer nowhere-zero 4-flows is nonempty and connected. Here two distinct flows are adjacent exactly when the nonzero support of their difference is a connected 2-regular subgraph.*

The argument does not prove integer connectivity for P_3 , nor does it establish the general modular-to-integer converse. The comparison made here is with the cited current versions and does not support a claim of priority.

Throughout, a cycle is a connected 2-regular subgraph. The cited papers formulate adjacency as equality away from some cycle. For two distinct flows this agrees with the exact-support convention in Theorem 1.1: their difference is a nonzero circulation supported on that cycle, and conservation at its degree-two vertices forces the signed difference to be a single nonzero constant around the whole cycle.

2 Coordinates for flows on an odd prism

Write the vertices of P_n as u_i, v_i for $i \in \mathbb{Z}/n\mathbb{Z}$. For the moment, orient the top and bottom rim edges from index i to index $i + 1$, and orient the spoke $u_i v_i$ from u_i to v_i . Write the corresponding flow values as a_i, b_i, r_i .

Lemma 2.1 (Prism coordinates). *Over $\mathbb{Z}$ or $\mathbb{Z}/4\mathbb{Z}$, an edge function in these coordinates is a flow if and only if, for some constant S ,*

$$r_i = a_{i-1} - a_i, \quad b_i = S - a_i \quad (i \in \mathbb{Z}/n\mathbb{Z}). \quad (1)$$

If n is odd, every nowhere-zero $\mathbb{Z}/4\mathbb{Z}$ -flow has $S = 0$ and is therefore equivalent to a proper cyclic word $c = (c_i)$ over $\{1, 2, 3\}$ via

$$a_i = c_i, \quad b_i = -c_i, \quad r_i = c_{i-1} - c_i \quad \text{in } \mathbb{Z}/4\mathbb{Z}. \quad (2)$$

Conversely, each such proper word gives a nowhere-zero modular flow and has an integer nowhere-zero 4-flow lift.

Proof. Conservation at u_i and v_i says, respectively,

$$a_i + r_i = a_{i-1}, \quad b_i = b_{i-1} + r_i.$$

These equations are equivalent to (1); conversely, (1) implies both conservation equations.

Now work in $\mathbb{Z}/4\mathbb{Z}$. If $S = 1, 2$, or 3 , the simultaneous requirements $a_i \neq 0$ and $S - a_i \neq 0$ restrict the possible values of a_i to, respectively,

$$\{2, 3\}, \quad \{1, 3\}, \quad \{1, 2\}.$$

Nonzero spokes require $a_{i-1} \neq a_i$, which would properly color the odd cycle with two colors. Hence $S = 0$. Equations (2) now identify modular flows with proper cyclic 3-color words.

For an integer lift, take the ordinary representatives $c_i \in \{1, 2, 3\}$, put $a_i = c_i$, $b_i = -c_i$, and $r_i = c_{i-1} - c_i$. Rim magnitudes are at most 3, and each spoke has magnitude 1 or 2 because the word is proper. $\square$

3 All modular edges are interval moves

A *proper cyclic interval* below means a nonempty set

$$I = \{s, s + 1, \dots, s + \ell - 1\} \subsetneq \mathbb{Z}/n\mathbb{Z}, \quad 1 \leq \ell \leq n - 1,$$

with indices interpreted cyclically.

Lemma 3.1 (Move classification). *Let c, c' be distinct proper cyclic words over $\{1, 2, 3\}$, regarded as modular flows by (2). Their difference support is a single cycle if and only if there are a proper cyclic interval I and a constant $\delta \in \{1, 2, 3\}$ such that*

$$c'_i = c_i + \delta \mathbf{1}_{i \in I} \pmod{4} \quad \text{for every } i. \quad (3)$$

In that case the support consists of the two rim paths indexed by I and the two boundary spokes at s and $s + \ell$.

Proof. Put $d_i = c'_i - c_i$ in $\mathbb{Z}/4\mathbb{Z}$. The top and bottom rim differences at index i are nonzero exactly when $d_i \neq 0$, while the difference on spoke i is nonzero exactly when $d_{i-1} \neq d_i$. At either vertex of index i , the degree in the support is therefore

$$[d_{i-1} \neq 0] + [d_i \neq 0] + [d_{i-1} \neq d_i]. \quad (4)$$

Two consecutive unequal nonzero d -values would make this degree 3. Thus each maximal cyclic block of nonzero entries is constant.

If every d_i is nonzero, there is one constant block and the support is the disjoint union of the two complete rims, so it is not connected. If there is a proper nonzero block, its two rim paths and boundary spokes form one cycle. Distinct maximal blocks yield disjoint cycles, even when their separating zero block has length one. Hence connectedness is equivalent to having exactly one proper nonzero block, which is precisely (3). The converse follows from the same description of the support. Since $1 \leq \ell \leq n - 1$ and $n \geq 3$, the boundary spokes are distinct. $\square$

4 Lifting every modular edge

Theorem 4.1 (Integer edge lifting). *Let $n \geq 3$ be odd. Every edge between nowhere-zero $\mathbb{Z}/4\mathbb{Z}$ -flows on P_n has endpoint lifts F, G that are adjacent integer nowhere-zero 4-flows. The integer and modular differences have exactly the same support, and F and G agree away from that cycle.*

Proof. By Lemma 3.1, write the modular move as (3). We construct integer words x, y , set

$$F = (x_i, -x_i, x_{i-1} - x_i)_{i \in \mathbb{Z}/n\mathbb{Z}}, \quad G = (y_i, -y_i, y_{i-1} - y_i)_{i \in \mathbb{Z}/n\mathbb{Z}}, \quad (5)$$

and use Lemma 2.1 for conservation.

If $\delta = 1$, a source color inside I is 1 or 2, since 3 would become 0. Take $x_i = c_i$ everywhere and $y_i = x_i + \mathbf{1}_{i \in I}$. Both x and y are the ordinary representatives of proper words, so every rim value lies in $\{1, 2, 3\}$ up to sign and every spoke has magnitude 1 or 2.

If $\delta = 3$, source colors inside I are 2 or 3. Take $x_i = c_i$ and $y_i = x_i - \mathbf{1}_{i \in I}$. The same bound and nonvanishing argument applies, and the update -1 represents 3 modulo 4.

Suppose $\delta = 2$. The source colors inside I are 1 and 3 and alternate whenever there is an internal adjacency. At either boundary, the unchanged exterior color must differ from both the source interior color and its target; these two colors are 1 and 3. Thus every exterior boundary color is 2. Define

$$x_i = \begin{cases} -1, & i \in I \text{ and } c_i = 3, \\ c_i, & \text{otherwise,} \end{cases} \quad y_i = x_i + 2\mathbf{1}_{i \in I}.$$

Inside I , the representatives 1, -1 become 3, 1. An internal spoke has value ± 2 before and after, and an exterior spoke is unchanged. At a boundary, in the order exterior minus interior, the only possibilities are

$$2 - 1 = 1 \mapsto 2 - 3 = -1, \quad 2 - (-1) = 3 \mapsto 2 - 1 = 1.$$

The opposite boundary order only negates these values. Hence all rim and spoke values before and after the move lie in $\{-3, -2, -1, 1, 2, 3\}$.

This boundary calculation also covers $\ell = 1$, when there are no internal spokes. If $\ell = n - 1$, the two exterior boundary neighbors are the same index; both conditions give it the consistent value 2, while the two boundary spokes remain distinct.

In the three cases, put $t = 1, -1, 2$, respectively. From (5), $G - F$ equals t on the top rim edges indexed by I , $-t$ on the corresponding bottom rim edges, $-t$ on the entering boundary spoke, t on the leaving boundary spoke, and 0 elsewhere. Its exact support is therefore the cycle from Lemma 3.1. The residues of F, G are the prescribed modular endpoints, completing the lift. $\square$

The theorem is existential at the level of residue fibers. It does not say that every integer representative can perform a prescribed modular move immediately. The next standard lemma supplies the paths within a fiber.

Lemma 4.2 (Residue fibers [2]). *Let f, g be integer nowhere-zero 4-flows on any finite graph, in one fixed orientation. If $f(e) \equiv g(e) \pmod{4}$ for every edge e , then a sequence of legal single-cycle updates joins f to g .*

Proof. Set $h = g - f$. Since every endpoint value has magnitude at most 3, each $h(e)$ belongs to $\{-4, 0, 4\}$. Direct every support edge in its fixed orientation when $h(e) = 4$ and in the reverse direction when $h(e) = -4$. Conservation of h makes this directed support balanced. It therefore decomposes into edge-disjoint directed simple cycles: follow unused arcs until a vertex repeats, remove the resulting cycle, and iterate.

Process these cycles one at a time, adding the signed value 4 on their edges. Each edge is changed at most once, directly from its value in f to its value in g . Thus every intermediate coordinate is one of its two valid endpoint values. Conservation is preserved, and each step has exactly one cycle as its difference support. The process terminates after at most $|E|$ steps. $\square$

5 Connectivity and an explicit procedure

Proof of Theorem 1.1. Let f, g be arbitrary integer nowhere-zero 4-flows on P_n , with n odd and $n \geq 5$, and reduce them modulo 4. Cranston et al.'s modular prism theorem [1, Theorem 3.4] supplies a finite modular path between the reductions. For each modular edge on this path, Theorem 4.1 supplies adjacent integer endpoints. Connect f to the first lifted endpoint inside its residue fiber using Lemma 4.2; after each lifted edge, use the same lemma to reach the chosen lift of the next edge. Finally connect to g in its fiber. If the modular path has length zero, the fiber lemma alone applies. This gives a finite legal integer path between arbitrary states.

For nonemptiness, take the cyclic word

$$1, 2, 1, 2, \dots, 1, 2, 3,$$

where the alternating prefix has length $n - 1$, and apply the integer lift in Lemma 2.1. Finally, reversing any initially chosen edge orientation and negating that edge coordinate in every state is an isomorphism of reconfiguration graphs. Thus the result holds for every fixed orientation. $\square$

The proof also describes a terminating transformation. Enumerate the at most 3^n cyclic words and their valid interval moves, use breadth-first search for a modular path, realize its edges with Theorem 4.1, and decompose each fiber difference into directed cycles as in Lemma 4.2. No complexity bound uniform in n is needed for the connectivity assertion.

6 Exact checks and scope

An exact-arithmetic checker was used to attack the sign and boundary claims. It enumerated all 36 valid local transition types; generated all interval moves for $n = 3, 5, 7, 9, 11$; compared the interval classification with definition-based all-pairs modular adjacency for $n \leq 9$; and tested fiber paths for all $360 + 3472 = 3832$ integer states in complete certificates for P_5 and P_7 . In those two cases the integer state graphs have, respectively, 360 vertices and 2300 edges, and 3472 vertices and 35602 edges, each with one component. These are finite falsification tests. The universal conclusion comes from Lemma 3.1, Theorem 4.1, Lemma 4.2, and the cited modular connectivity theorem.

Exact finite certification also appears elsewhere in discrete mathematics; for a broadly adjacent example using different enumerative methods, see [3]. That work contains no result about flows, prisms, or the theorem proved here and is cited only as computational-methodology context.

The edge-lifting theorem includes $n = 3$, but the modular connectivity input does not yield integer connectivity for the triangular prism. Even prisms are outside the stated theorem. The result is a resolution for the odd prism family, not for the general question of whether modular flow connectivity implies integer flow connectivity.

Computational and AI assistance disclosure. OpenAI Codex assisted with proof exploration, exact finite-case checking, adversarial review of interval boundaries and residue-fiber concatenation, and preparation of the manuscript. The checker used exact integer arithmetic and was rerun against the reported cases. Computational output was treated only as corroboration; it was not used to infer the theorem for unbounded n . The manuscript records the accepted mathematical scope and distinguishes the universal argument from its finite checks. These computational and AI checks do not constitute external peer review; no such review is claimed here.

References

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